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Research paper

Baicalein disrupts TGF-β-induced EMT in pancreatic cancer by FTO-dependent m6A demethylation of ZEB1

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ABSTRACT

Pancreatic ductal adenocarcinoma (PDAC) is a highly aggressive malignancy associated with poor prognosis. Baicalein, a flavonoid extracted from the roots of Scutellaria baicalensis, traditionally used in Chinese medicine, has demonstrated potential in inhibiting cancer development and progression. However, its mechanism of action remains poorly understood, particularly regarding epigenetic gene regulation through m6A RNA methylation. In this study, three human PDAC cell lines and one nonmalignant cell line were employed. The effects of baicalein were examined using multiple assays, including RT-qPCR, MeRIP-qPCR, Western blotting, spheroid formation, RNA stability, and MTT, to evaluate cellular functions and m6A regulation. Baicalein significantly reduced cell viability, migration, invasion, and colony formation. It also downregulated FTO, an enzyme critical for m6A RNA demethylation. Knockdown of FTO replicated the effects of baicalein, underscoring its oncogenic role in PDAC. Bioinformatic analysis identified ZEB1—a key transcription factor in epithelial-to-mesenchymal transition—as an m6A-modified target regulated by FTO. Both baicalein treatment and FTO knockdown enhanced m6A modification and decreased *ZEB1* mRNA stability, thereby suppressing stemness-related features. Rescue experiments further confirmed that baicalein disrupts the TGF-β/FTO/ZEB1 signaling axis, highlighting its therapeutic potential in PDAC. This study offers fundamental insights for the development of novel therapeutic strategies targeting PDAC.

1. Introduction

PDAC accounts for approximately 90 % of pancreatic cancer cases [1] and remains a major global health burden due to its aggressive behavior and poor prognosis [2]. According to GLOBOCAN 2022 data from the International Agency for Research on Cancer (IARC), there were 510,992 new cases and 467,409 deaths from pancreatic cancer worldwide, ranking it as the 12th most common malignancy and the 6th leading cause of cancer-related mortality [3]. Despite medical advances, pancreatic cancer continues to have a low 5-year survival rate of 13 % [4], primarily due to delayed diagnosis. The incidence and mortality rates of pancreatic cancer are increasing [5], and it is projected to become the second leading cause of cancer-related deaths in the United States by 2030 [6].

Epithelial–mesenchymal transition (EMT) is a biological process in which epithelial cells acquire mesenchymal characteristics, gaining enhanced motility and invasive capabilities [7]. This process is regulated by multiple signaling pathways and stimuli, with TGF-β serving as a major inducer [8]. EMT-associated transcription factors (EMT-TFs), including SNAIL, SLUG, ZEB1, ZEB2, TWIST1, and TWIST2, promote EMT by downregulating epithelial markers (e.g., E-cadherin) and upregulating mesenchymal markers (e.g., N-cadherin and Vimentin).

N6-methyladenosine (m6A) is the most prevalent, abundant, and evolutionarily conserved internal co-transcriptional modification in eukaryotic RNA, particularly in higher organisms [9]. It is dynamically regulated by specific enzymes categorized as writers, erasers, and readers. "Writers" such as METTL3 and METTL14 catalyze methylation of adenosine residues; "erasers" such as FTO and ALKBH5 remove these

Abbreviations: ALKBH5, alkB homologue 5; EMT, epithelial-mesenchymal transition; EMT-TFs, EMT transcription factors; FTO, fat mass and obesity-associated protein; GEO, gene expression omnibus; GTEx, genotype-tissue expression; KEGG, kyoto encyclopedia of genes and genomes; m6A, N6-Methyladenosine; MeRIP, m6A RNA immunoprecipitation; METTL3, methyltransferase-like 3; METTL14, methyltransferase-like 14; PDAC, pancreatic ductal adenocarcinoma; TCGA, tissue cancer genome atlas; TCM, traditional Chinese medicine; TGF, transforming growth factor; ZEB1, zinc finger *E*-box-binding homeobox 1.

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modifications; and "readers" like YTHDF1 and YTHDF2 bind to m6A-modified transcripts to regulate RNA stability, translation, and splicing [9]. By modulating m6A levels in oncogenes and tumor suppressor genes, m6A modification plays a pivotal role in tumor biology, including proliferation, metastasis, drug resistance, and immune response [10].

Baicalein, a natural flavonoid derived from the root of Scutellaria baicalensis Georgi, exhibits antioxidant, antiviral, antibacterial, and anticancer activities [11]. *In vitro* and preclinical studies have demonstrated its anticancer effects in various malignancies, including prostate [12], liver [13], pancreatic [14], gastric [15], colorectal [16], and breast cancer [17]. Baicalein inhibits cell proliferation and metastasis and induces apoptosis and autophagy by modulating signaling pathways such as ERK/MAPK, mTOR, TGF-β, and PI3K/Akt/NF-κB [18]. Despite its broad anticancer activities, the precise mechanisms underlying baicalein's effects remain unclear, especially in the context of m6A regulation. Further investigation into its impact on m6A modification is essential for improving our mechanistic understanding and developing more effective anticancer therapies.

2. Materials and methods

2.1. Cell lines

The established human PDAC cell lines PANC-1, BxPC-3, MIA-PaCa2, AsPC-1, and the non-malignant hTERT-immortalized epithelial cell line CRL-4023, derived from human pancreatic ducts, were obtained from the American Type Culture Collection (ATCC, Manassas, VA, USA). All cell lines, except CRL-4023, were cultured in high-glucose DMEM (Sigma, Deisenhofen, Germany) supplemented with 10 % fetal bovine serum (FBS) and 25 mmol/L HEPES. CRL-4023 cells were maintained in a medium consisting of 75 % glucose-free DMEM supplemented with 2 mM L-glutamine, 1.5 g/L sodium bicarbonate, and 25 % M3 Base Medium (Incell Corporation LLC, San Antonio, TX, USA), further supplemented with 5 % FBS and 10 ng/mL puromycin. All cells were cultured in flasks and incubated at 37 °C in a humidified atmosphere containing 5 % CO_2. Mycoplasma contamination was routinely ruled out using the PlasmoTest™ assay (InvivoGen, San Diego, USA), performed monthly. All cell lines were recently authenticated by single nucleotide polymorphism (SNP) analysis conducted by Multiplexion (Heidelberg, Germany).

2.2. Reagents

Baicalein (Cat. No. 465119, Sigma-Aldrich) was dissolved in DMSO to prepare a 100 mM stock solution. Recombinant human TGF-β1 protein (Cat. No. 7754-BH-005, R&D Systems) was reconstituted in sterile 4 nM HCl to a final concentration of 100 μg/mL. Actinomycin D (Cat. No. A7592, Invitrogen, Thermo Fisher Scientific) was dissolved in DMSO to yield a 4 mg/mL stock solution. All reagents were aliquoted and stored at −20 °C, and each aliquot was used only once immediately after thawing. The final concentration of solvents in cell culture media was maintained at 0.1 % (*v*/v) or less.

2.3. RT-qPCR

Total cellular RNA was extracted using the RNeasy Mini Kit (QIAGEN, Hilden, Germany) according to the manufacturer's protocol. Reverse transcription was performed using the High-Capacity RNA-to-cDNA Kit (Thermo Fisher Scientific, Hilden, Germany). Quantitative real-time PCR (RT-qPCR) was carried out using PowerUp SYBR Green Master Mix (Thermo Fisher Scientific, Hilden, Germany), with *GAPDH* used as an internal control gene. Reactions were conducted on the StepOne Real-Time PCR System (Applied Biosystems, Darmstadt, Germany) under the following cycling conditions: initial denaturation at 95 °C for 2 min, followed by 40 cycles of denaturation at 95 °C for 15 s, annealing at 56 °C for 15 s, and extension at 72 °C for 1 min. Primer pairs were synthesized by Eurofins Genomics (Ebersberg, Germany), and their sequences are listed in Suppl. Table S1.

2.4. Western blot analyses

Cells were lysed in RIPA buffer (Sigma-Aldrich, Steinheim, Germany) according to the standard protocol. Protein concentrations were determined using the BCA Protein Assay Kit (Thermo Fisher Scientific, USA). For Western blot analysis, 40–60 μg of total protein per lane was separated by SDS–PAGE and transferred to Immobilon-FL PVDF membranes (Merck) using a wet transfer system. The following primary antibodies were used: rabbit monoclonal antibodies against GAPDH, β-Actin, E-Cadherin, N-Cadherin, Vimentin, Smad2, phospho-Smad2, CD44, and c-Myc (all from Cell Signaling Technology, Danvers, MA, USA); and mouse monoclonal antibodies against FTO and ZEB1 (Proteintech, Germany). Membranes were incubated with primary antibodies overnight at 4 °C, followed by incubation with IRDye 700 or 800CW-conjugated goat anti-rabbit or goat anti-mouse IgG secondary antibodies (LI-COR Biosciences, Bad Homburg, Germany) for 1 h at room temperature. Protein bands were visualized using the Odyssey CLx Imaging System (LI-COR Biosciences).

2.5. Small interference RNA and plasmid DNA transfection

Cells were transfected using Lipofectamine 2000 (Thermo Fisher Scientific, Dreieich, Germany) according to the manufacturer's instructions. For transient knockdown of FTO, FlexiTube siRNA and AllStars Negative Control siRNA (QIAGEN, Hilden, Germany) were used. For transient overexpression, cells were transfected with a human FTO cDNA ORF clone (Cat. No. HG12125-UT, SinoBiological) or a negative control vector (Cat. No. CV011, SinoBiological).

2.6. Cell viability assay

To evaluate the inhibitory effects of baicalein, PDAC cells (8000 cells/well) were seeded into 96-well plates and treated with increasing concentrations of baicalein for 24, 48, and 72 h. Control cells were treated with an equivalent volume of DMSO. Prior to measurement, MTT solution (5 mg/mL; 20 μL/well) was added and incubated for 4 h at 37 °C in the dark. After removing the supernatant, formazan crystals were dissolved in 200 μL of DMSO and mixed thoroughly. Absorbance was measured at 570 nm using a microplate reader. Cell viability was determined based on optical density (OD) values. Dose–response curves were plotted and IC_{50} values were calculated to determine the optimal concentrations of baicalein. For post-transfection viability analysis, transfected cells were cultured for 24, 48, 72, and 96 h after seeding, and OD values were measured as described above. All experiments were performed in triplicate.

2.7. Wound healing assay

Cells were seeded into 6-well plates and cultured at 37 °C until reaching approximately 90 % confluence. Following 24 h of serum starvation in medium without fetal calf serum (FCS), a linear scratch was created using a 10 μL pipette tip. Cells were gently washed with PBS to remove debris and then incubated in 2 mL of serum-free medium. The initial wound area (0 h) was imaged using a Leica microscope. Wound closure was subsequently recorded at defined time points: 6, 12, and 24 h for baicalein-treated cells, and 24 h for FTO-knockdown cells, using ×100 magnification. Wound areas were analyzed using ImageJ software, and the percentage of wound closure was calculated relative to the initial wound size.

2.8. Transwell invasion assay

Transwell invasion assays were performed using 24-well invasion chambers with 6.5 mm diameter inserts and 8 μm pore size membranes (Costar, Corning Incorporated, NY, USA). The inserts were coated with Matrigel® matrix (Corning, Kaiserslautern, Germany) diluted to 50 μg/mL in coating buffer (0.01 M Tris, pH 8.0, 0.7 % NaCl) at 4 °C. A volume of 100 μL of the diluted Matrigel was added to each insert and incubated at 37 °C for 2 h. Residual liquid was then removed prior to seeding. Cells from each treatment group were harvested by trypsinization, and 1×10^5 cells were suspended in serum-free medium and seeded into the upper chamber. The lower chambers were filled with DMEM supplemented with 20 % FCS to serve as a chemoattractant. After 24 h of incubation, cells were fixed with 4 % paraformaldehyde and stained with 0.05 % Coomassie Blue. Non-invading cells on the upper surface of the membrane were removed using a cotton swab. Invaded cells on the underside of the membrane were visualized under a microscope (Nikon, Düsseldorf, Germany). Five random fields per insert were selected, and the number of invaded cells was quantified using ImageJ software to calculate the invasion index.

2.9. Colony formation assay

Cells from each treatment group were trypsinized and reseeded at a low density of 600 cells per well in triplicate. The plates were incubated at 37 °C for 2 weeks without a medium change until visible colonies were observed under a microscope. After incubation, the cells were rinsed with 2 mL PBS, fixed with 2 mL of 4 % paraformaldehyde for 10 min, and subsequently treated with 2 mL of 70 % ethanol for an additional 10 min. Cells were then stained with 0.05 % Coomassie Blue, washed with water, and air-dried overnight. Colonies containing at least 50 cells were counted using a microscope (Nikon, Düsseldorf, Germany).

2.10. Spheroid formation assay

Cells from each treatment group were trypsinized and re-seeded into 24-well ultra-low attachment plates (Corning™ Costar™ Ultra-Low Attachment Microplates). Cells were cultured in 500 μL of complete NeuroCult® NS-A Proliferation Medium (Human) per well. The medium consisted of NeuroCult NS-A basal serum-free medium (Human) (StemCell Technologies, Vancouver, BC, Canada), supplemented with 2 mg/mL heparin (StemCell Technologies), 20 ng/mL human epidermal growth factor (hEGF; R&D Systems, Wiesbaden Nordenstadt, Germany), 10 ng/mL human basic fibroblast growth factor (hFGF-b; PeproTech GmbH, Hamburg, Germany), and NeuroCult NS-A Proliferation Supplements (Human) (StemCell Technologies). After 5 days of incubation at 37 °C, spheroids were imaged under a microscope at ×200 magnification. The number and diameter of spheroids were quantified using ImageJ software.

2.11. Quantification of m6A modifications

Global m6A RNA methylation levels were quantified using the EpiQuik™ m6A RNA Methylation Quantification Kit (Colorimetric) (Epigentek, New York, USA), according to the manufacturer's instructions. Briefly, 200 ng of RNA was added to assay wells containing the binding solution. Capture antibody, detection antibody, and enhancer solutions were sequentially added at the recommended dilutions. Colour development was initiated with developer solution and terminated with stop solution. Absorbance was measured at 450 nm, and m6A levels were calculated based on the standard curve.

2.12. Methylated RNA immunoprecipitation and RT-qPCR (MeRIP-qPCR)

Total RNA was extracted using the RNeasy Mini Kit (QIAGEN, Hilden, Germany) as previously described. Methylated RNA immunoprecipitation (MeRIP) was performed using the m6A RNA Enrichment Kit (Epigentek, Farmingdale, USA) following the manufacturer's instructions. In brief, 20 μg of total RNA was incubated with anti-m6A antibody and affinity beads for 90 min to form RNA–antibody–bead complexes. After washing and purification, enriched RNA fragments were eluted and subjected to RT-qPCR. m6A enrichment was calculated by normalizing the signal from immunoprecipitated RNA to the input RNA.

2.13. RNA stability assay

Following 48 h of baicalein treatment or siRNA transfection, transcription was inhibited by adding 5 μg/mL actinomycin D to the culture medium. Cells were then collected at 0, 2, 4, and 6 h post-treatment. Total RNA was extracted using the RNeasy Mini Kit (QIAGEN, Hilden, Germany) and analyzed by RT-qPCR to assess RNA degradation over time.

2.14. Tumor xenotransplantation on fertilized chicken eggs

Fertilized eggs from genetically identical Lohmann Brown hybrid chickens were obtained from a local ecological hatchery (Geflügelzucht Hockenberger, Eppingen, Germany). After disinfection with warm 70 % ethanol, the eggs were placed vertically in a sterile incubator at 37.8 °C, with the pointed end facing downward. On embryonic day 4, a small window was created in the eggshell to assess embryo viability. Viable embryos were then returned to the incubator and maintained in a horizontal position. On embryonic day 9, BxPC-3 cells (1×10^6) were resuspended in 50 μL of DMEM mixed with 50 % Matrigel. Sterile coverslips with a central hole were placed onto the chorioallantoic membrane (CAM). On embryonic day 14, 100 μL of 50 μM baicalein solution or 100 μL of saline was directly injected into CAM vessels supplying the xenograft tumors. On embryonic day 18, the embryos were humanely euthanized, and tumors were harvested. Tumor volume was measured using calipers under a Nikon camera, applying the formula: $V = 4/3 \times \pi \times a \times b \times c$, where a, b, and c represent the three radii.

2.15. Immunohistochemical staining

Immunohistochemistry was performed on 3–5 μm paraffin-embedded tissue sections. Sections were deparaffinized in xylene, rehydrated through a graded ethanol series, and subjected to antigen retrieval in 10 mM sodium citrate buffer (pH 6.0) at high temperature. Endogenous peroxidase activity was blocked using 3 % hydrogen peroxide for 10 min, followed by blocking with 5 % normal goat serum for 1 h at room temperature. Primary antibodies were incubated overnight at 4 °C, including mouse monoclonal antibodies against Ki-67 (Dako, Agilent), E-cadherin (Cell Signaling Technology), and ZEB1 (Proteintech, Germany). After PBS washing, secondary antibody incubation was performed using the Histofine Simple Stain MAX PO (M) kit (Nichirei Biosciences Inc.) for 1 h. Staining was developed using DAB as the chromogen. Sections were then counterstained with hematoxylin, dehydrated through ethanol, cleared in xylene, and mounted with Entellan (Merck, Germany). Positive staining was quantified using ImageJ software (https://imagej.net/Downloads).

2.16. In silico analysis

Bioinformatics analyses were performed using a range of online tools and publicly available databases. Tumor versus normal differential gene expression analysis was conducted using GEPIA (Gene Expression Profiling Interactive Analysis; http://gepia.cancer-pku.cn), based on data from TCGA (The Cancer Genome Atlas; https://www.cancer.gov/ccg/research/genome-sequencing/tcga) and GTEx (Genotype-Tissue Expression; https://gtexportal.org/home/). Kaplan–Meier survival

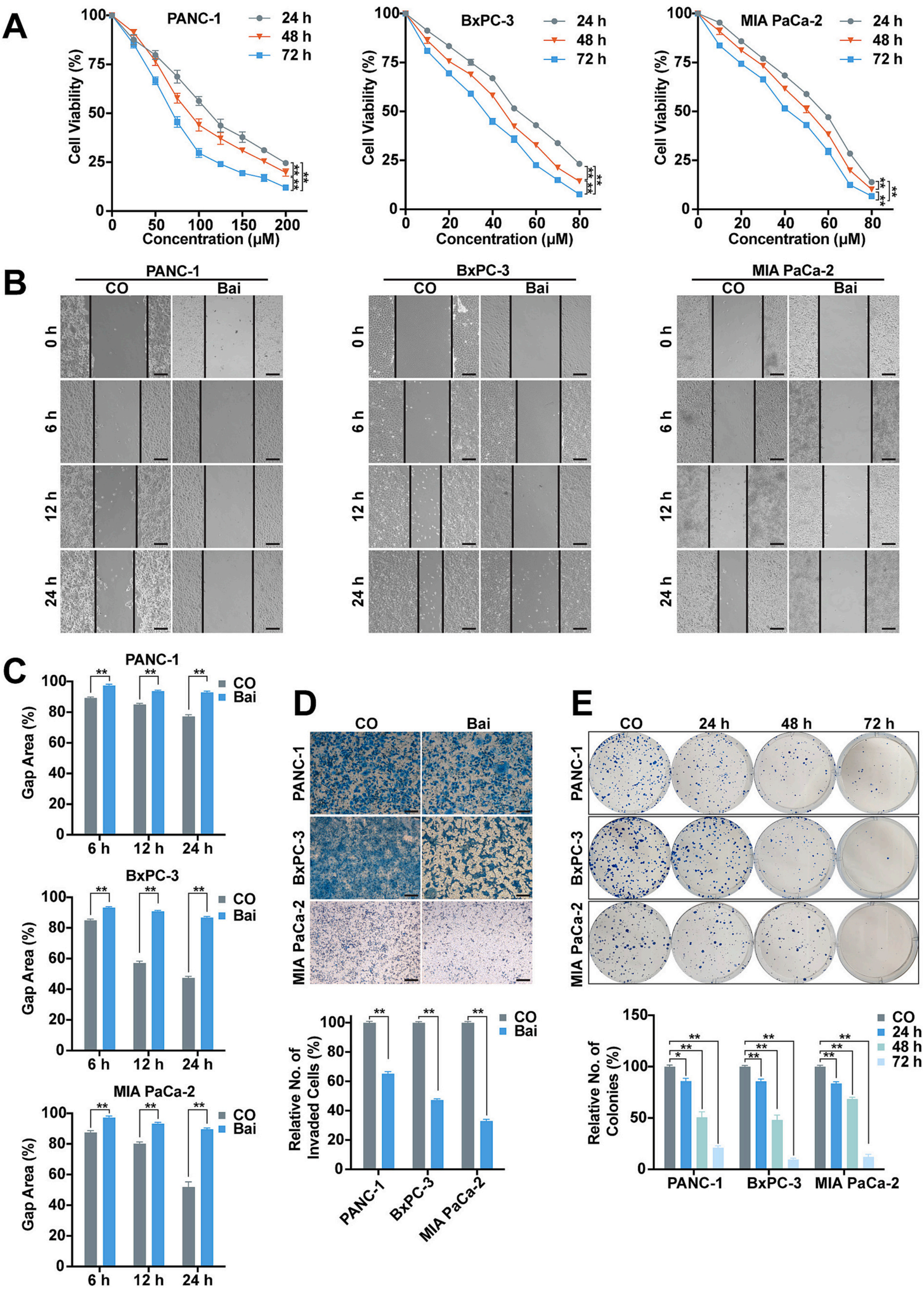

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Fig. 1. Baicalein inhibits tumor progression features.
(A) Cell viability was assessed by MTT assay in PDAC cells treated with various concentrations of baicalein for 24, 48, and 72 h. Statistical analysis: two-way ANOVA followed by Tukey's post hoc test ($n = 3$).
(B, C) Scratch (wound healing) assay was performed on PDAC cells pre-grown to 90 % confluency and treated with or without baicalein for 24 h. Wound closure was monitored microscopically at 0, 6, 12, and 24 h (×100 magnification). ImageJ analyzed the gap area proportion relative to the initial area. Statistical analysis: Student's *t*-test.
(D) For the transwell invasion assay, PDAC cells were treated with baicalein for 48 h. A total of 1×10^5 cells/well were seeded into 24-well invasion chambers and incubated for 24 h. Cells that invaded through the membrane were fixed with 4 % PFA and stained with 0.05 % Coomassie Blue. Non-invading cells were removed, and invaded cells on the underside of the membrane were visualized under a microscope (×100 magnification). Five random fields per group were analyzed using ImageJ. Statistical analysis: Student's t-test (n = 3).
(E) Colony formation assay: cells were pretreated with baicalein for 24, 48, or 72 h, then seeded at 600 cells/well in 6-well plates. After 14 days, colonies were fixed with 3.7 % PFA, stained with 0.05 % Coomassie Blue, and counted under a dissecting microscope. Plating efficiency was normalized to untreated. Statistical analysis: one-way ANOVA followed by Dunnett's multiple comparisons test.
Baicalein concentrations: 50 μM for BxPC-3 and MIA PaCa-2; 100 μM for PANC-1.
CO: Control; Bai: Baicalein. Scale bar: 50 μm. Data are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

curves were generated using the Kaplan–Meier Plotter (https://pancreas.kmplot.com). RNA-seq expression data from the GSE57711 dataset were obtained from the GEO database (Gene Expression Omnibus; https://www.ncbi.nlm.nih.gov/geo/). Differentially expressed genes (DEGs) were identified using the "limma" package in RStudio (https://www.rstudio.com). The RM2Target database (http://rm2target.canceromics.org/) was used to identify m6A-modified genes. Correlation analysis in PDAC was performed using TCGA RNA-seq data in RStudio. Functional enrichment analyses were conducted using KEGG (Kyoto Encyclopedia of Genes and Genomes; http://www.genome.jp/kegg) and GO (Gene Ontology; http://geneontology.org/), and visualized with the "clusterProfiler" and "ggplot2" packages in RStudio. TIMER 2.0 (http://timer.cistrome.org/) was used to assess correlations between FTO and EMT-TFs in PDAC. The SRAMP tool (http://www.cuilab.cn/sramp) was used to predict potential RNA m6A modification sites.

2.17. Statistical analysis

Quantitative data are presented as mean ± standard error of the mean (SEM) from at least three independent experiments, unless stated otherwise. Comparisons between two groups were performed using either Student's *t*-test or the Mann–Whitney *U* test. For comparisons involving more than two groups, one-way ANOVA followed by Dunnett's or Tukey's multiple comparisons test was applied. Two-way ANOVA was used for experiments involving two independent variables, followed by Tukey's post hoc test. A *P*-value <0.05 was considered statistically significant. Significance levels are indicated as follows: $P < 0.05$ (*), $P < 0.01$ (**).

3. Results

3.1. Baicalein inhibits tumor progression features

To comprehensively evaluate the effects of baicalein on pancreatic cancer progression, PANC-1, BxPC-3, and MIA PaCa-2 cells were treated with baicalein at concentrations ranging from 0 to 200 μM. Cell viability was assessed at 0, 24, 48, and 72 h post-treatment using the MTT assay. A significant, dose- and time-dependent decrease in cell viability was observed across all three PDAC cell lines (Fig. 1A). The half-maximal inhibitory concentration (IC_{50}) values were calculated for each cell line (Suppl. Table S2) and used in all following experiments. In addition, baicalein treatment significantly inhibited cell migration (Fig. 1B, C), invasion (Fig. 1D), and colony formation (Fig. 1E) in all tested PDAC cell lines.

3.2. Baicalein suppresses highly expressed FTO in PDAC

To determine whether m6A RNA methylation is involved in baicalein-induced tumor suppression, we first measured the global m6A levels in PDAC cells following baicalein treatment using a colorimetric ELISA-like assay. The results showed a marked increase in total m6A levels upon baicalein exposure (Fig. 2A), suggesting that m6A modification may play a role in baicalein-mediated anti-tumor effects. To further explore the regulatory mechanisms, we assessed the mRNA expression of four key m6A regulators— *METTL3*, *METTL14*, *FTO*, and *ALKBH5*—by RT-qPCR. Among them, *FTO* expression was consistently downregulated across all PDAC cell lines after baicalein treatment, while the expression of the other regulators remained largely unaffected (Fig. 2B). Consistently, Western blot analysis confirmed a time-dependent decrease in FTO protein levels at 24, 48, and 72 h post-treatment (Fig. 2C, D). Moreover, FTO protein expression was markedly elevated in all four examined PDAC cell lines compared to the nonmalignant pancreatic ductal epithelial cell line CRL-4023 (Fig. 2E, F). Bioinformatic analysis using TCGA and GTEx datasets further corroborated these findings, revealing significantly higher *FTO* expression in PDAC tissues compared to normal pancreatic tissues (Fig. 2G). To evaluate the clinical relevance of FTO, Kaplan–Meier survival analysis was performed using the KM Plotter (https://pancreas.kmplot.com/). The analysis showed that high *FTO* expression was associated with significantly worse overall survival ($P = 0.0372$) and disease-free survival ($P = 1.32 \times 10^{-7}$) in PDAC patients (Fig. 2H).

3.3. Knockdown of FTO inhibits tumor progression features

To investigate the role of FTO in PDAC progression, we transfected PANC-1, BxPC-3, and MIA PaCa-2 cells with four distinct siRNA constructs targeting FTO (siFTO_5, siFTO_6, siFTO_7, and siFTO_8), along with a nonsense siRNA control (NC). RT-qPCR (Suppl. Fig. S1) and Western blot analysis (Fig. 3A, B) confirmed that siFTO_5, siFTO_6, and siFTO_7 effectively suppressed FTO expression across all three cell lines, with siFTO_5 exhibiting the most robust knockdown efficiency. Therefore, siFTO_5 was selected for use in subsequent experiments and referred to as siFTO. As expected, knockdown of FTO resulted in a significant increase in global m6A levels in PDAC cells, as determined by m6A quantification assay (Fig. 3C), indicating enhanced m6A RNA methylation. Consistently, MTT assays demonstrated a significant reduction in cell viability following FTO silencing compared to the NC (Fig. 3D). Furthermore, FTO knockdown markedly impaired tumor progression features in PDAC cells, including decreased migratory capacity (Fig. 3E, F), invasive ability (Fig. 3G, H), and colony-forming potential (Fig. 3I, J).

3.4. ZEB1 is a potential m6A target of FTO in baicalein-mediated tumor suppression

To elucidate the regulatory mechanism of FTO under baicalein treatment, we first analyzed the publicly available RNA-seq dataset GSE221059 from the GEO database to identify genes differentially expressed following baicalein exposure. A total of 2338 differentially

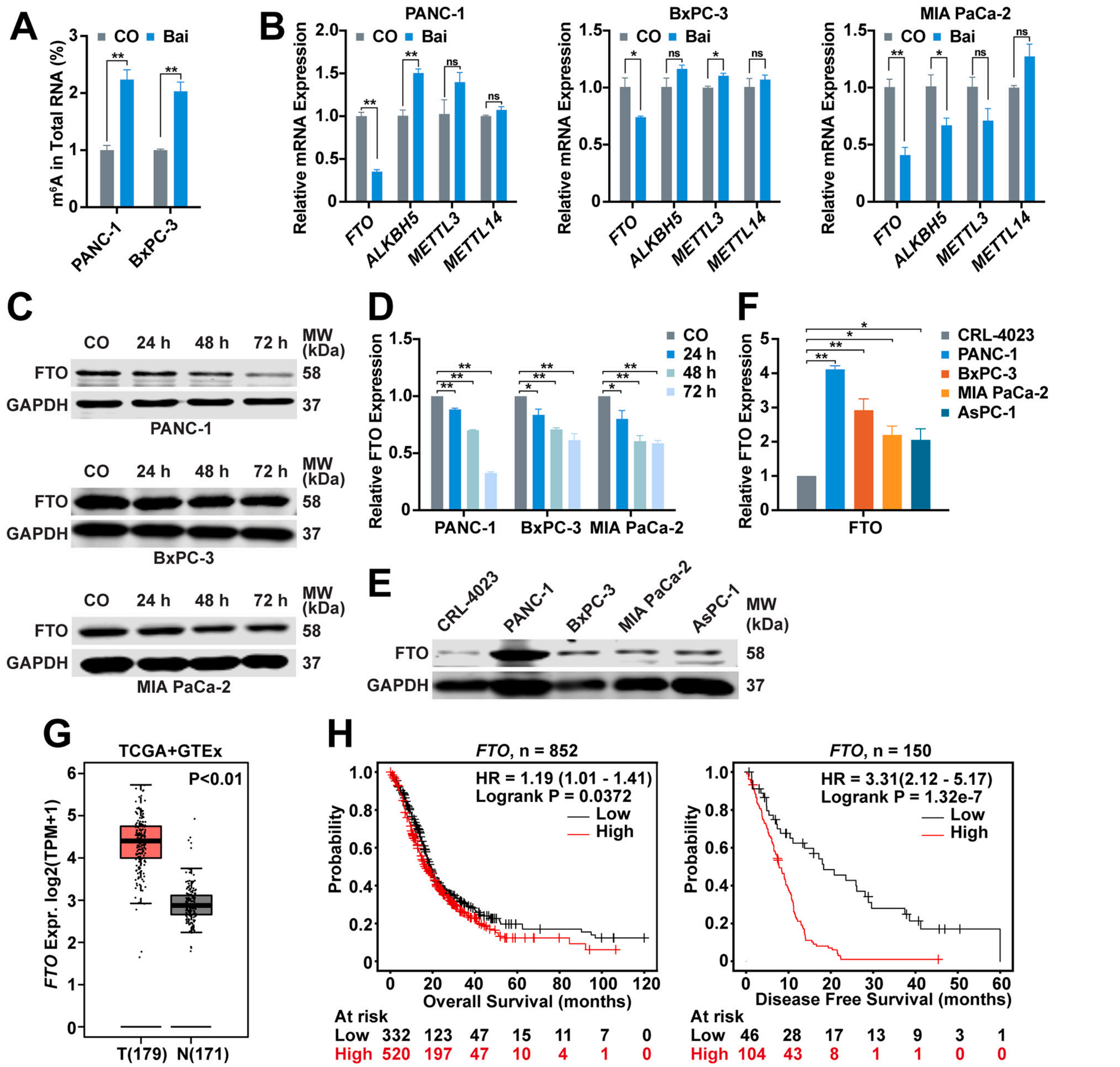

Fig. 2. Baicalein suppresses highly expressed FTO in PDAC.

(A) Total m6A levels in PDAC cells following baicalein treatment were quantified. The m6A level of the control group was normalized to 1. Statistical analysis: Student's *t*-test ($n = 3$).

(B) mRNA expression levels of *FTO*, *ALKBH5*, *METTL3*, and *METTL14* were assessed by RT-qPCR in PDAC cells treated with baicalein for 48 h and normalized to *GAPDH*. Expression in the control group was set to 1. Statistical analysis: Student's t-test ($n = 3$).

(C, D) Western blot analysis of FTO protein expression in PDAC cells treated with baicalein for 24, 48, and 72 h. GAPDH was used as a loading control. Statistical analysis: two-way ANOVA followed by Tukey's test (n = 3).

(E, F) Western blot analysis of FTO expression in four PDAC cell lines (PANC-1, BxPC-3, MIA PaCa-2, AsPC-1) and non-malignant pancreatic ductal epithelial cells (CRL-4023). Statistical analysis: one-way ANOVA followed by Dunnett's test ($n = 3$).

(G) Box plots comparing *FTO* mRNA expression levels between PDAC tissues (T; $n = 179$) and normal pancreatic tissues (N; $n = 171$) using data from TCGA and GTEx databases via the GEPIA platform.

(H) Kaplan–Meier survival curves (overall survival (OS), $n = 852$; disease-free survival (DFS), $n = 150$) based on *FTO* mRNA expression using the KM Plotter (GEO dataset). High (red) and low (black) FTO expression groups were stratified using the optimal cutoff.

Baicalein concentrations: 50 μM for BxPC-3 and MIA PaCa-2; 100 μM for PANC-1.

CO: Control; Bai: Baicalein; HR: Hazard ratio. Values are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

expressed genes (DEGs) were identified between baicalein-treated and control groups using the criteria $|\log(\text{fold change})| > 1$ and $P < 0.05$. In parallel, correlation analysis using TCGA PDAC data revealed 1414 genes significantly correlated with FTO expression (correlation coefficient ≥ 0.5). In addition, target prediction from the RM2Target database identified 8305 RNAs that potentially interact with FTO in an m6A-dependent manner. Intersection of these three gene sets yielded 422 common genes, which are likely to be regulated by FTO via m6A modification under baicalein treatment (Fig. 4A).

GO enrichment analysis of the 422 overlapping genes revealed the top 10 enriched terms across biological processes, molecular functions, and cellular components categories. Notably, terms such as "cellular

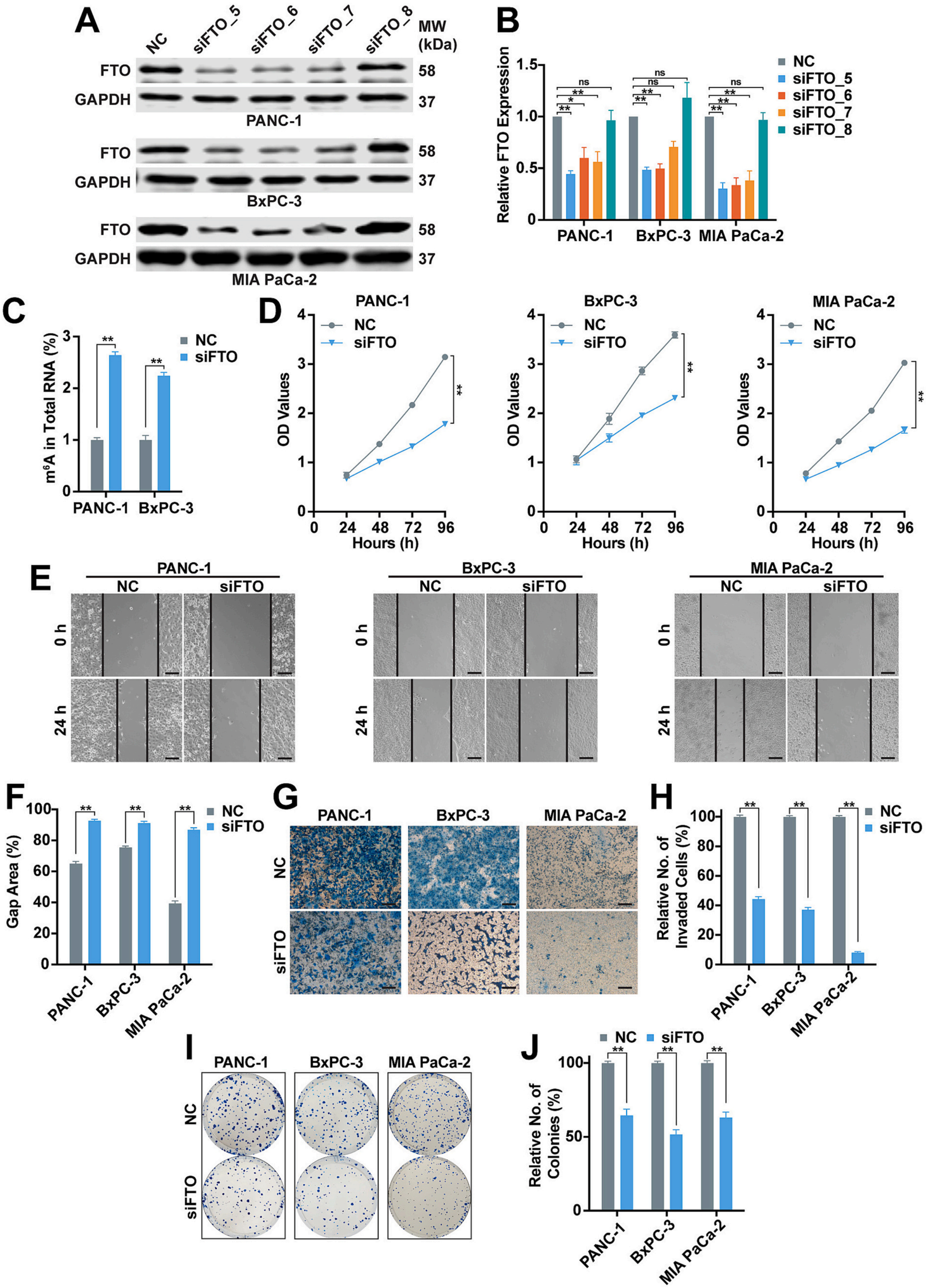

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Fig. 3. Knockdown of FTO inhibits tumor progression features.
(A, B) PDAC cells were transfected with siRNAs targeting FTO or NC. After 48 h, FTO protein expression was assessed by Western blot, with GAPDH as the loading control. Statistical analysis: one-way ANOVA followed by Dunnett's test ($n = 3$).
(C) Total m6A levels in PDAC cells following siFTO transfection. m6A levels in the NC group were normalized to 1. Statistical analysis: Student's *t*-test ($n = 3$).
(D) Cell viability was assessed by MTT assay at 24, 48, and 72 h post-transfection. Statistical analysis: two-way ANOVA ($n = 3$).
(E, F) Wound healing assay was performed 24 h post-lipotransfection. When cells reached ~90 % confluence, scratches were made using a 10 μL pipette tip. Wound areas were imaged at 0 and 24 h under ×100 magnification. ImageJ quantified wound closure relative to the initial gap. Statistical analysis: Student's *t*-test.
(G, H) Transwell invasion assay was conducted 24 h post-transfection. A total of 1×10^5 cells/well were seeded, and invaded cells were quantified after 24 h. Statistical analysis: Student's *t*-test.
(I, J) Colony formation assay was performed 24 h post-transfection. Statistical analysis: Student's t-test.
Baicalein concentrations: 50 μM for BxPC-3 and MIA PaCa-2; 100 μM for PANC-1.
NC: Negative control. Scale bar: 50 μm. Values are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

response to transforming growth factor beta stimulus," "DNA-binding transcription activator activity specific to RNA polymerase II" and "endoplasmic reticulum lumen" were among the most enriched respectively (Fig. 4B). KEGG pathway enrichment analysis further identified the TGF-β signaling pathway as the most significantly enriched among these genes (Fig. 4C). Given these findings, particularly the frequent appearance of extracellular matrix-related processes in GO results, we focused our attention on the role of TGF-β-induced EMT.

Using TIMER2, we examined the correlation between *FTO* and key EMT-TFs (*SNAI1*, *SNAI2*, *TWIST1*, *TWIST2*, *ZEB1*, *ZEB2*) across multiple cancer types. *ZEB1* exhibited the strongest correlation with *FTO* in PDAC (Fig. 4D; Suppl. Fig. S2). To evaluate the m6A modification potential of *ZEB1* mRNA, we used SRAMP, a sequence-based m6A site predictor. SRAMP identified seven high- or very high-confidence m6A modification sites within the *ZEB1* transcript (Fig. 4E).

To experimentally verify whether *ZEB1* is a direct m6A-regulated target of FTO, we performed RT-qPCR analysis. Baicalein treatment or FTO knockdown significantly reduced *ZEB1* mRNA expression in both PANC-1 and BxPC-3 cells (Fig. 4F, G). MeRIP-qPCR assays further revealed a significant increase in m6A enrichment on *ZEB1* mRNA following baicalein treatment or FTO knockdown (Fig. 4H, I). Considering the role of FTO in regulating mRNA stability, we conducted RNA stability assays and observed that both baicalein treatment and FTO silencing substantially reduced the half-life of *ZEB1* mRNA (Fig. 4J, K).

3.5. *Baicalein inhibits TGF-β-induced EMT as mimicked by FTO knockdown*

To investigate the effect of baicalein on TGF-β-induced EMT in PDAC cells, cells were treated with baicalein, TGF-β1 (10 ng/mL), or both for 48 h. Western blot analysis revealed that baicalein alone did not significantly alter Smad2 or phosphorylated Smad2 (p-Smad2) expression compared to the control. However, baicalein effectively reversed the TGF-β1-induced upregulation of both Smad2 and p-Smad2 (Fig. 5A–C).

Morphological examination showed that PANC-1 cells treated with TGF-β1 acquired an elongated, spindle-shaped, fibroblast-like appearance—typical of cells undergoing EMT—while baicalein-treated cells retained a cobblestone-like epithelial morphology. Notably, co-treatment with baicalein reversed the TGF-β1-induced morphological changes (Fig. 5D). Consistent with these observations, Western blot analysis demonstrated that baicalein significantly attenuated TGF-β1-induced EMT marker alterations (Fig. 5E, F). Specifically, baicalein prevented the downregulation of the epithelial marker E-cadherin and the upregulation of mesenchymal markers N-cadherin, Vimentin, and ZEB1. Moreover, baicalein treatment reversed the TGF-β1-induced increase in FTO protein levels, implicating FTO in the regulation of TGF-β-induced EMT.

To further examine whether FTO mediates TGF-β1-induced EMT, PDAC cells were transfected with siFTO and subsequently treated with TGF-β1. Morphologically, siFTO-transfected cells retained an epithelial phenotype and reversed the fibroblast-like changes induced by TGF-β1 (Fig. 5G). Western blot assays confirmed that FTO knockdown countered the TGF-β1-induced decrease in E-cadherin and increase in N-cadherin, Vimentin, and ZEB1 (Fig. 5H, I). These results suggest that FTO plays a mediating role in TGF-β-induced EMT.

3.6. *Baicalein inhibits TGF-β/FTO/ZEB1 signaling and thereby tumor progression*

To determine whether baicalein exerts its anti-tumor effects in PDAC cells via FTO, PANC-1 cells were transfected with an FTO overexpression plasmid and subsequently treated with baicalein. The efficiency of FTO overexpression was confirmed by Western blot analysis (Fig. 6A). Functional assays revealed that FTO overexpression promoted cell viability and colony formation, and notably reversed the inhibitory effects of baicalein on both phenotypes (Fig. 6B–D). Western blot analysis further demonstrated that FTO overexpression significantly counteracted baicalein-induced molecular changes associated with EMT. Specifically, it reversed the upregulation of E-cadherin and restored the expression of N-cadherin, Vimentin, and ZEB1 (Fig. 6E). These results support the conclusion that baicalein inhibits tumor progression by suppressing the TGF-β/FTO/ZEB1 signaling axis.

To validate the molecular mechanism, we next examined the effect of FTO overexpression on m6A RNA methylation. A colorimetric m6A quantification assay revealed a significant reduction in global m6A levels following FTO plasmid transfection (Fig. 6F). In parallel, RT-qPCR analysis showed that FTO overexpression led to a significant increase in *ZEB1* mRNA expression (Fig. 6G). MeRIP-qPCR assays demonstrated decreased m6A enrichment on *ZEB1* mRNA in cells overexpressing FTO (Fig. 6H). Additionally, RNA stability assays confirmed that *ZEB1* mRNA exhibited a prolonged half-life upon FTO overexpression (Fig. 6I), suggesting that FTO enhances *ZEB1* expression by reducing its m6A methylation and stabilizing its transcript.

To evaluate the effect of baicalein on tumor growth *in vivo*, a tumor xenotransplantation assay was performed using fertilized chicken eggs (Suppl. Fig. S3A). In our previous studies, we demonstrated that pancreatic tumors rapidly engraft in the chorioallantoic membrane (CAM) of fertilized chicken eggs, and that the resulting xenografts closely resemble primary patient tumors in terms of morphology, expression profiles, progression markers, and pancreatic ductal markers [19]. Based on these advantages, the CAM model was selected for this study. Baicalein-treated xenografts exhibited significantly reduced tumor volumes compared to the control group (Suppl. Fig. S3B, C). Immunohistochemical analysis of xenograft tissue sections revealed a marked decrease in the proliferation marker Ki-67 expression following baicalein treatment. In addition, baicalein-treated xenografts showed increased E-cadherin expression and decreased ZEB1 expression levels, indicating suppression of EMT (Suppl. Fig. S3D, E). Collectively, these results demonstrate that baicalein effectively inhibits tumor growth and EMT progression *in vivo*.

3.7. *Baicalein suppresses tumor stemness via downregulation of FTO*

During EMT epithelial cells often acquire stem cell-like properties, including enhanced self-renewal capacity and elevated expression of

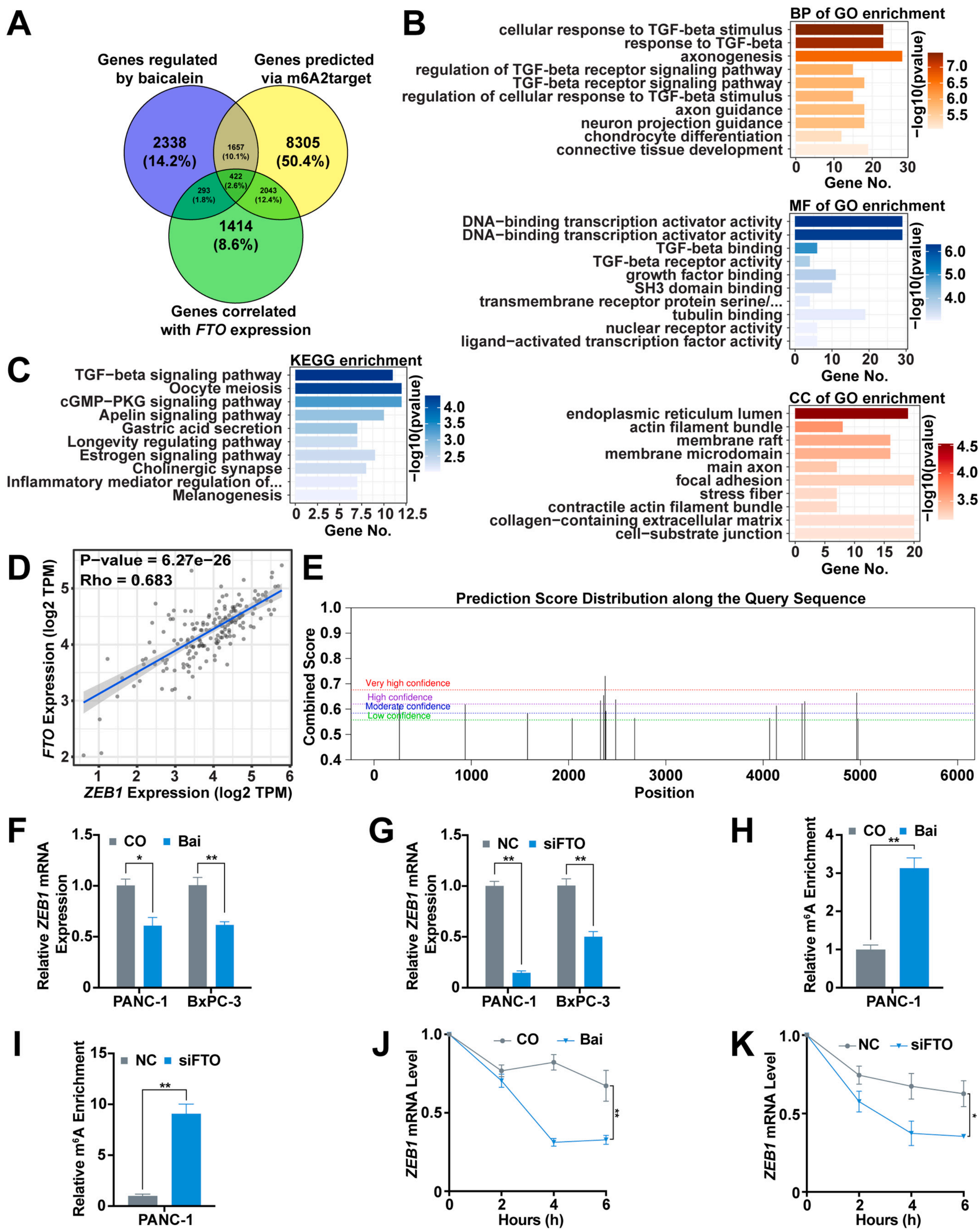

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Fig. 4. ZEB1 is a potential m6A target of FTO in baicalein-mediated tumor suppression.
(A) Venn diagram showing 422 overlapping genes identified from three sources: GSE221059 (baicalein-regulated genes), m6A2Target (FTO-mediated m6A targets), and TCGA (genes correlated with FTO expression).
(B) Gene Ontology (GO) enrichment analysis of the common genes, presenting the top 10 significantly enriched terms across three categories: biological process (BP), molecular function (MF), and cellular component (CC).
(C) KEGG pathway enrichment analysis of the overlapping genes showing the top 10 significantly enriched pathways.
(D) Correlation analysis of *FTO* and *ZEB1* mRNA expression in 179 PDAC samples using TCGA data (analyzed via TIMER2). Spearman's correlation coefficient: $R = 0.683$, $P = 6.27 \times 10^{-26}$.
(E) Prediction of m6A modification sites in the ZEB1 transcript using the SRAMP tool, showing the distribution of predicted m6A motifs.
(F, G) qRT-PCR analysis of *ZEB1* mRNA expression in PDAC cells treated with baicalein or transfected with siFTO. Statistical analysis: Student's *t*-test ($n = 3$).
(H, I) MeRIP-qPCR showing increased m6A modification of *ZEB1* mRNA after baicalein treatment or FTO knockdown in PANC-1 cells. Statistical analysis: Student's t-test (n = 3).
(J, K) RNA stability assays demonstrating reduced *ZEB1* mRNA stability following baicalein treatment or FTO knockdown in PANC-1 cells. Statistical analysis: two-way ANOVA (n = 3).
Baicalein concentrations: 50 μM for BxPC-3; 100 μM for PANC-1.
CO: Control; Bai: Baicalein; NC: Negative control. Data are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

stemness markers. To evaluate the effect of baicalein on self-renewal ability, we performed spheroid formation assays in PDAC cells treated with baicalein or transfected with siFTO. Both treatments significantly reduced the number and size of tumor spheres formed, indicating impaired self-renewal capacity (Fig. 7A, B).

To determine whether baicalein suppresses stemness through FTO regulation, PANC-1 cells were transfected with an FTO overexpression plasmid followed by baicalein treatment. FTO overexpression significantly increased both the number and size of spheroids and effectively reversed the inhibitory effects of baicalein (Fig. 7C–E), suggesting that baicalein impairs stem-like features via FTO downregulation. We further examined the expression of stemness-related proteins CD44 and c-Myc in PANC-1 cells. Western blot analysis revealed that baicalein treatment reduced CD44 and c-Myc expression, whereas FTO overexpression restored their levels (Fig. 7F, G). These findings indicate that baicalein suppresses stem cell marker expression through inhibition of FTO.

Based on the collective results, we propose a working model in which baicalein suppresses FTO expression by inhibiting the TGF-β signaling pathway, resulting in increased m6A modification of *ZEB1* mRNA. The elevated m6A levels destabilize *ZEB1* transcripts, thereby reducing ZEB1 expression. This, in turn, attenuates EMT progression and impairs tumor invasiveness and metastasis (Fig. 7H).

4. Discussion

In this study, we uncovered a novel regulatory pathway through which baicalein inhibits the progression of PDAC. We first confirmed the inhibitory effects of baicalein on PDAC. Notably, baicalein treatment resulted in a marked reduction in the expression of FTO, a key m6A RNA demethylase. Functional assays revealed that FTO knockdown suppressed malignant phenotypes, whereas FTO overexpression partially reversed the inhibitory effects of baicalein. Through bioinformatic analyses, we hypothesized that the TGF-β signaling pathway may be involved in the regulation of FTO by baicalein. Subsequent rescue experiments supported this hypothesis by demonstrating that: (i) baicalein inhibits TGF-β-induced EMT; (ii) FTO mediates TGF-β-induced EMT; and (iii) baicalein suppresses EMT via FTO downregulation. Furthermore, we identified *ZEB1* as a potential m6A-modified target of FTO in the context of TGF-β-induced EMT. Mechanistically, FTO promoted ZEB1 expression by stabilizing its mRNA through m6A demethylation. Collectively, our findings indicate that inhibition of the TGF-β/FTO/ZEB1 axis underlies the anti-tumor effects of baicalein in PDAC.

Our findings, for the first time, demonstrate that baicalein effectively inhibits TGF-β-induced EMT in PDAC cells, consistent with its reported anti-EMT effects in other cancer types [15,20,21]. SMAD4 is inactivated in approximately 50–55 % of PDAC cases [22], and several studies have demonstrated that TGF-β can promote tumor progression in the later stages of carcinogenesis through both SMAD4-dependent and SMAD4-independent mechanisms [23,24]. Given the high prevalence of SMAD4 loss in PDAC and the existence of compensatory TGF-β signaling mechanisms, we employed three PDAC cell lines—including the SMAD4-deficient BxPC-3 cell line—to investigate the effects of baicalein under different SMAD4 genetic backgrounds. Our findings revealed that baicalein effectively inhibits TGF-β-mediated EMT in PDAC cells through both SMAD4-dependent and SMAD4-independent mechanisms.

Several studies have reported that baicalein can inhibit the TGF-β signaling pathway either by directly targeting TGF-β receptor 1 kinase activity or by reducing TGF-β1 production. For instance, Zhang et al. demonstrated that baicalein selectively binds to the ATP-binding pocket of activin receptor-like kinase 5 (ALK5), thereby directly inhibiting its kinase activity [25]. In a separate study, Zheng et al. showed that baicalin, a structurally related flavonoid derivative, significantly reduced serum and tissue TGF-β1 levels in a renal fibrosis model [26]. Additionally, SUMOylation of Smad2 has been identified as a crucial post-translational modification that enhances its phosphorylation and promotes TGF-β-induced EMT [27]. A recent study by P. Zhang et al. demonstrated that baicalin upregulates the expression of SENP1 (Sentrin/SUMO-specific protease 1) [28], which mediates the deSUMOylation of target proteins. Based on these findings, it is reasonable to hypothesize that baicalein may exert similar effects by promoting the deSUMOylation of Smad2, thereby reducing its phosphorylation, stability, and overall protein expression. However, this proposed mechanism remains speculative and requires further experimental validation.

In addition to its anti-tumor potential, baicalein has also been shown to attenuate fibrotic processes. Previous studies have reported that baicalein mitigates renal fibrosis by suppressing TGF-β-induced proliferation, extracellular matrix accumulation, collagen synthesis, and SMAD3 phosphorylation in renal interstitial fibroblast cells [29,30]. Taken together, these findings highlight the therapeutic promise of baicalein not only as an anti-cancer agent but also as an inhibitor of TGF-β-driven fibrotic pathways that are highly relevant in PDAC.

In this study, we are the first to report that baicalein increases global m6A level in PDAC cells by downregulating FTO. Our results are in line with emerging evidence showing that flavonoids modulate m6A regulatory machinery across various cancer types. For instance, quercetin has been shown to synergize with cisplatin in cervical cancer cells by inhibiting proliferation, migration, and invasion through the suppression of METTL3 expression [31]. In nasopharyngeal carcinoma, baicalin increases METTL3 and METTL14 expression while concurrently reducing FTO and ALKBH5 expression [32]. Similarly, epigallocatechin gallate has been shown to modulate cyclin A2 and CDK2 expression in an m6A-dependent manner by suppressing FTO and enhancing YTHDF2 expression [33]. These studies collectively support our findings and further underscore the role of baicalein in regulating m6A in cancer.

Initially identified for its association with obesity, FTO gained recognition for its role in regulating energy homeostasis and body weight. More recently, its function as an m6A RNA demethylase has attracted considerable attention [34]. In our study, we observed

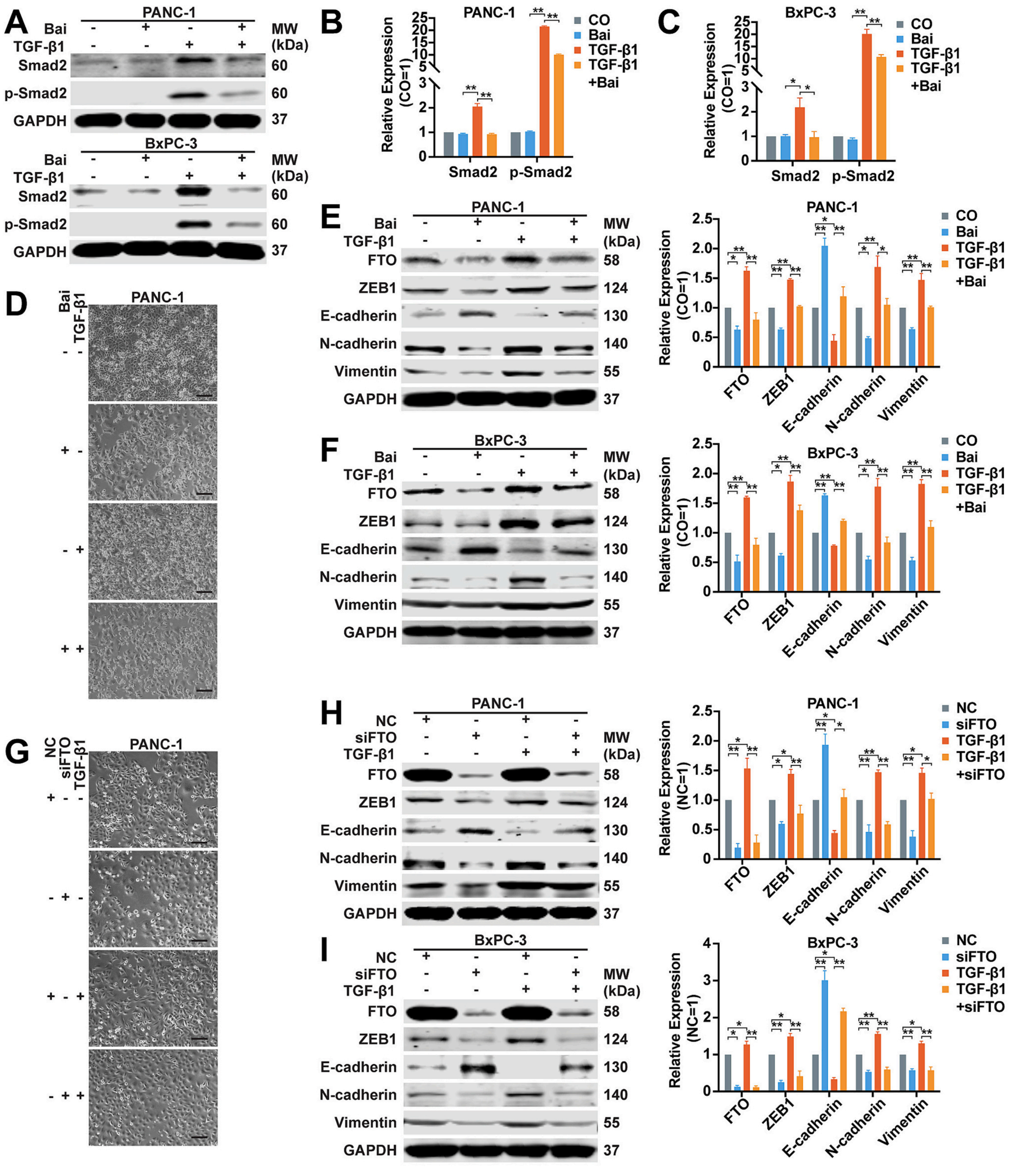

Fig. 5. Baicalein inhibits TGF-β-induced EMT as mimicked by FTO knockdown.

(A-C) Western blot analysis of Smad2 and phosphorylated Smad2 (p-Smad2) expression in PDAC cells treated with baicalein, TGF-β1 (10 ng/mL), or both for 48 h. Statistical analysis: one-way ANOVA followed by Tukey's test (n = 3).

(D) Representative images of PANC-1 cell morphology after treatment with baicalein, TGF-β1 (10 ng/mL), or both for 96 h.

(E, F) Western blot analysis of EMT-related markers (ZEB1, E-cadherin, N-cadherin, and Vimentin) in PDAC cells treated with baicalein, TGF-β1, or both for 48 h. Statistical analysis: one-way ANOVA followed by Tukey's test (n = 3).

(G) Representative images of PANC-1 cell morphology following transfection with siFTO or NC and subsequent treatment with or without TGF-β1 (10 ng/mL) for 96 h.

(H, I) Western blot analysis of EMT-related markers in PDAC cells transfected with siFTO or NC and treated with or without TGF-β1 (10 ng/mL) for 48 h. Statistical analysis: one-way ANOVA followed by Tukey's test (n = 3).

Baicalein concentrations: 50 μM for BxPC-3; 100 μM for PANC-1.

CO: Control; Bai: Baicalein; NC: Negative control. Scale bar: 50 μm. Data are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

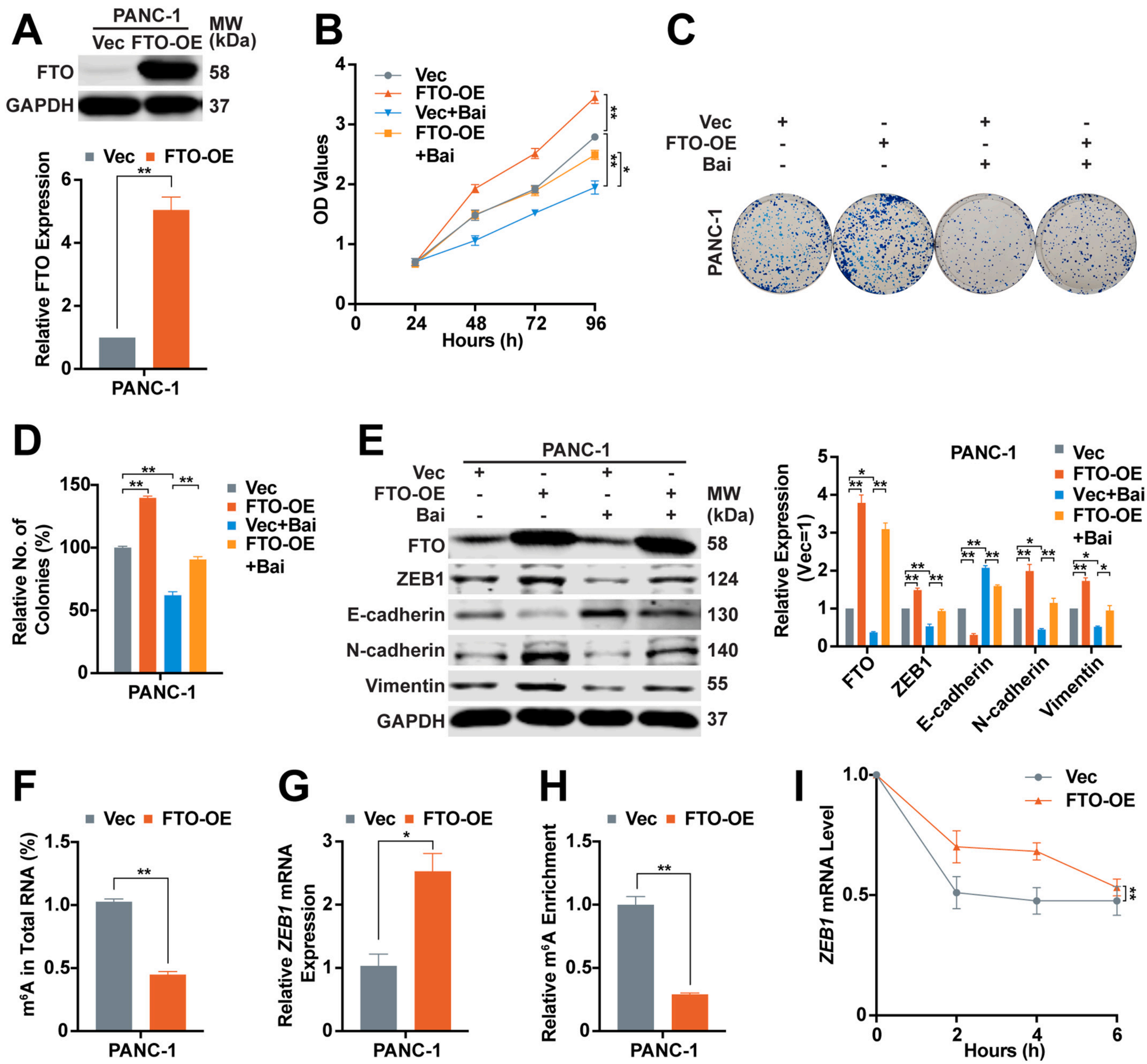

Fig. 6. Baicalein inhibits TGF-β/FTO/ZEB1 signaling and thereby tumor progression.
(A) PANC-1 cells were transfected with either a negative control vector (Vec) or an FTO overexpression plasmid (FTO-OE). After 48 h, FTO protein expression was evaluated by Western blot. Statistical analysis: Student's t-test (n = 3).
(B) Transfected cells were treated with or without baicalein for 48 h. Cell viability was assessed using the MTT assay at 24, 48, and 72 h post-treatment. Statistical analysis: two-way ANOVA followed by Tukey's test (n = 3).
(C, D) Colony formation assay of PANC-1 cells transfected with Vec or FTO-OE, followed by baicalein treatment or not for 48 h. Statistical analysis: one-way ANOVA followed by Tukey's test (n = 3).
(E) Western blot analysis of EMT-related markers (ZEB1, E-cadherin, N-cadherin, and Vimentin) following baicalein treatment in FTO-overexpressing or control PANC-1 cells. Statistical analysis: one-way ANOVA followed by Tukey's test (n = 3).
(F) Total m6A levels following FTO overexpression in PANC-1 cells. The m6A level in the Vec group was normalized to 1. Statistical analysis: Student's t-test (n = 3).
(G) qRT-PCR analysis of *ZEB1* mRNA expression following FTO overexpression. Statistical analysis: Student's t-test (n = 3).
(H) MeRIP-qPCR showing reduced m6A modification in *ZEB1* mRNA upon FTO overexpression. Statistical analysis: Student's t-test (n = 3).
(I) RNA stability assay indicating increased *ZEB1* mRNA stability in FTO-overexpressing PANC-1 cells. Statistical analysis: two-way ANOVA (n = 3).
Baicalein concentration: 100 μM for PANC-1.
Vec: Negative control vector; FTO-OE: FTO overexpression. Data are presented as mean ± SEM. P < 0.05 (*), P < 0.01 (**).

elevated FTO expression in PDAC cell, and functional assays revealed that FTO facilitates key malignant phenotypes in PDAC cells. These findings are consistent with previous studies that have established the oncogenic role of FTO across various cancer types— including lung cancer [35], breast cancer [36], melanoma [37], and pancreatic cancer [38]—where FTO significantly contributes to tumorigenesis and cancer progression. Furthermore, we demonstrated that FTO mediates TGF-β-induced EMT, in line with earlier research showing that FTO inhibition

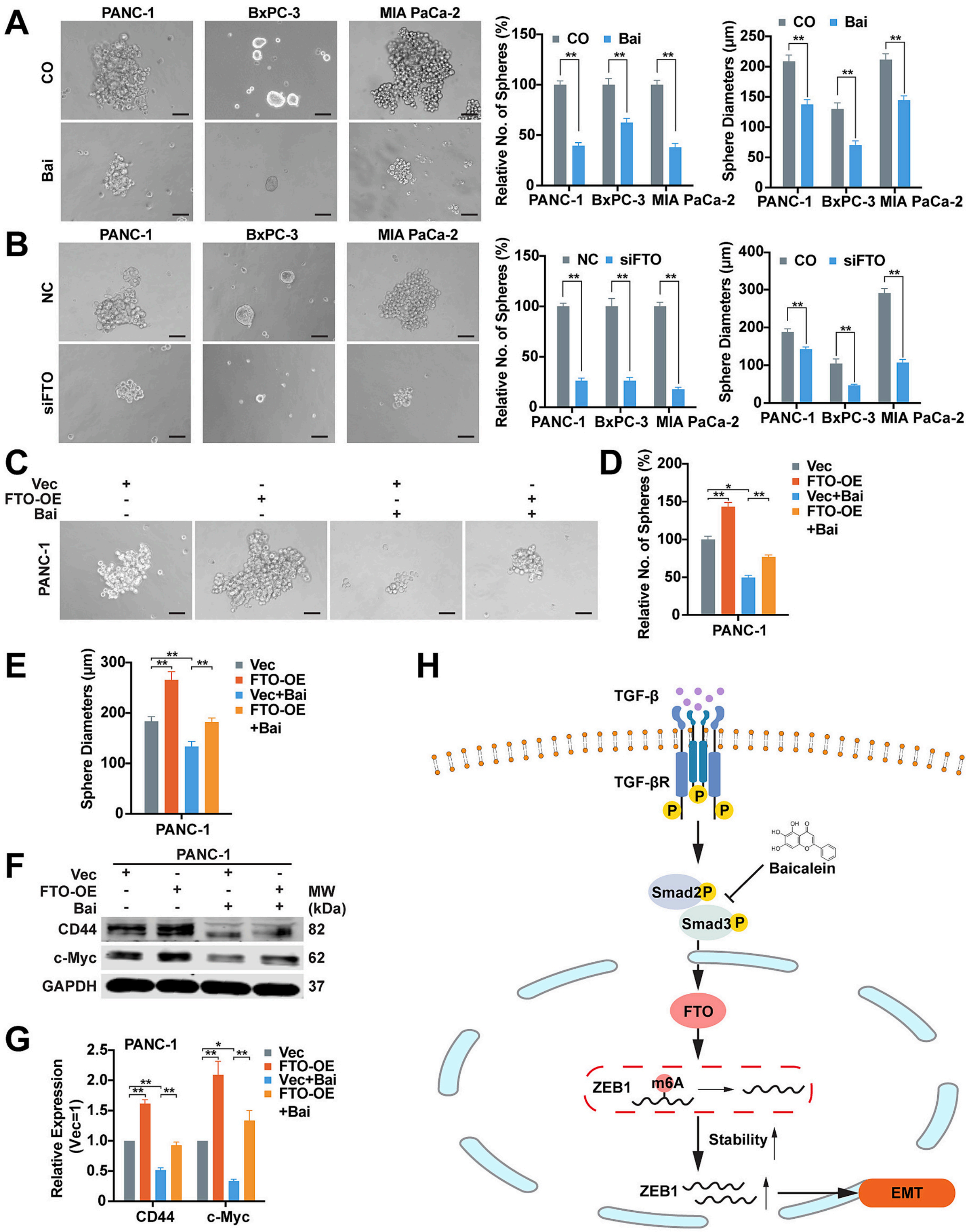

Fig. 7. Baicalein suppresses tumor stemness via downregulation of FTO.
(A) Spheroid formation assay showing that baicalein significantly reduced both the number and size of spheres formed by PANC-1 cells. Statistical analysis: Student's t-test or Mann–Whitney test.
(B) FTO knockdown also significantly reduced sphere number and size in PANC-1 cells. Statistical analysis: Student's t-test.
(C–E) Rescue experiment showing the effect of FTO overexpression on baicalein-suppressed spheroid formation. Statistical analysis: one-way ANOVA followed by Tukey's test.
(F, G) Western blot analysis showing the effect of FTO overexpression on baicalein-induced inhibition of stemness markers CD44 and c-Myc. Statistical analysis: one-way ANOVA followed by Tukey's test.
(H) Schematic diagram illustrating the proposed mechanism: baicalein inhibits the TGF-β signaling pathway, leading to reduced FTO expression and increased m6A modification of *ZEB1* mRNA. This destabilizes *ZEB1* mRNA and reduces ZEB1 expression, thereby attenuating TGF-β-induced EMT and suppressing tumor metastasis and invasion.
Baicalein concentrations: 50 μM for BxPC-3 and MIA PaCa-2; 100 μM for PANC-1.
Scale bar: 25 μm. Vec: Negative control vector; FTO-OE: FTO overexpression. Data are presented as mean ± SEM. $P < 0.05$ (*), $P < 0.01$ (**).

impairs cellular mobility and invasiveness, thereby reversing EMT [39]. Additionally, suppression of FTO has been associated with reduced tumorsphere formation and downregulation of cancer stem cell (CSC) markers [39], consistent with our findings. The role of FTO in fibrotic diseases has also been reported, with studies showing its upregulation in tubular epithelial cells exposed to TGF-β1 and in mouse models of renal interstitial fibrosis induced by unilateral ureteral obstruction (UUO) [40,41]. Silencing FTO attenuates TGF-β1- and UUO-induced EMT and inflammatory responses, further supporting FTO's involvement in TGF-β-driven fibrotic processes. Considering these findings, along with the documented ability of baicalein to mitigate renal fibrosis via inhibition of TGF-β-induced EMT, our results underscore the critical role of FTO in mediating baicalein's effects. Notably, our rescue experiments provide novel evidence that suppression of FTO is a key mechanism by which baicalein exerts its anti-tumor activity, particularly by inhibiting TGF-β-induced EMT and cancer stemness in PDAC.

Given that m6A is the most prevalent endogenous RNA modification in eukaryotic cells, it is plausible that EMT-associated transcription factors are also regulated through m6A-dependent mechanisms. Indeed, several studies have demonstrated m6A-mediated regulation of key EMT drivers across multiple cancer types. For instance, in ovarian [42,43], pancreatic [44] and nasopharyngeal carcinoma [45], the expression of *Snail* is modulated by m6A regulators such as FTO, ALKBH5, and METTL3. Similarly, in breast cancer, *ZEB1* expression is regulated in an m6A-dependent manner by FTO and YTHDF2 [46,47]. Moreover, IGF2BP2, an m6A reader, has been shown to promote lymphatic metastasis and EMT in head and neck squamous carcinoma by stabilizing *Slug* mRNA [48]. These findings collectively highlight the pivotal role of m6A methylation in the post-transcriptional regulation of EMT transcription factors. In our study, we identified *ZEB1* as a novel m6A-modified target of FTO in PDAC. Specifically, we demonstrated that FTO promotes *ZEB1* mRNA stability by reducing its m6A methylation level, thereby facilitating EMT. This finding aligns with previous reports indicating that FTO-mediated stabilization of *ZEB1* transcripts contributes to chemoresistance and EMT in breast cancer [46]. While we conducted a rescue experiment involving baicalein and FTO and demonstrated that FTO directly regulates *ZEB1* in an m6A-dependent manner, we acknowledge that including a specific rescue assay targeting ZEB1 directly would offer stronger mechanistic evidence.

Through a combination of extensive cellular experiments and comprehensive bioinformatic analyses, we delineated a novel m6A-dependent regulatory mechanism by which baicalein modulates the TGF-β/FTO/ZEB1 axis in PDAC. Rescue experiments further confirmed the functional relevance of this pathway, establishing a new axis of molecular regulation in PDAC progression. However, we acknowledge that several limitations of our study warrant further investigation. First, although our data demonstrated that baicalein reduces FTO expression by inhibiting the TGF-β signaling pathway, we cannot exclude the possibility that baicalein directly targets FTO. Previous studies employing fluorescence quenching assays have shown that flavonoids such as quercetin exhibit high binding affinity for FTO via hydrophobic interactions and hydrogen bonding [49]. Given the structural similarity, baicalein may also directly bind to FTO. Future structural studies, including X-ray crystallography or cryo-electron microscopy, are necessary to clarify this potential interaction. Second, while we employed MeRIP-qPCR to demonstrate that FTO directly regulates *ZEB1* in an m6A-dependent manner, additional assays are needed to further characterize this interaction at the molecular level. In particular, dual-luciferase reporter assays or mutational analyses could help to identify the specific m6A-binding sites involved in the regulation of *ZEB1* by FTO. Third, we acknowledge that the chicken embryo is not a mammalian system; therefore, tumor xenotransplantation in mice would provide a more physiologically relevant *in vivo* model.

Our identification of baicalein as an inhibitor of the key m6A demethylase FTO in PDAC provides valuable insights for future therapeutic strategies. Compared to conventional FTO or TGF-β inhibitors, baicalein is a naturally derived compound with superior biocompatibility [50]. In addition, baicalein exerts multi-targeted effects by modulating multiple signaling pathways and holds promise for use in combination with other chemotherapeutic agents [50,51]. Subsequent studies may focus on the development of structurally optimized baicalein derivatives with enhanced potency and reduced off-target effects as potential FTO-targeted agents. Furthermore, our elucidation of the regulatory mechanism by which baicalein modulates the TGF-β/FTO/ZEB1 axis offers a novel conceptual framework for drug development targeting this pathway in PDAC and potentially other malignancies.

5. Conclusion

In conclusion, our findings demonstrate that baicalein inhibits PDAC progression by downregulating FTO expression, thereby suppressing TGF-β-induced EMT through m6A-dependent regulation of *ZEB1* mRNA. These results highlight the therapeutic potential of targeting the TGF-β/FTO/ZEB1 signaling axis as a novel approach for PDAC treatment.

CRediT authorship contribution statement

Lian Zhao: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gong Chen:** Investigation. **Dan Li:** Investigation. **Kangtao Wang:** Investigation. **Michael Schaefer:** Software, Methodology. **Ingrid Herr:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Data curation, Conceptualization. **Bin Yan:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Consent for publication

All authors agree to the publication of this manuscript. This manuscript has not been published and is not under consideration for publication elsewhere.

Ethics statement

Does not apply.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work we used ChatGPT in order to achieve linguistic refinement. We have thoroughly checked the accuracy of the language improvement suggestions provided by ChatGPT and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that the project was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbamcr.2025.119969.

Data availability

The datasets supporting the conclusions of this article are included within the article, and its supplemental files are thus available. Additional files are available upon request and with legitimate interest.

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